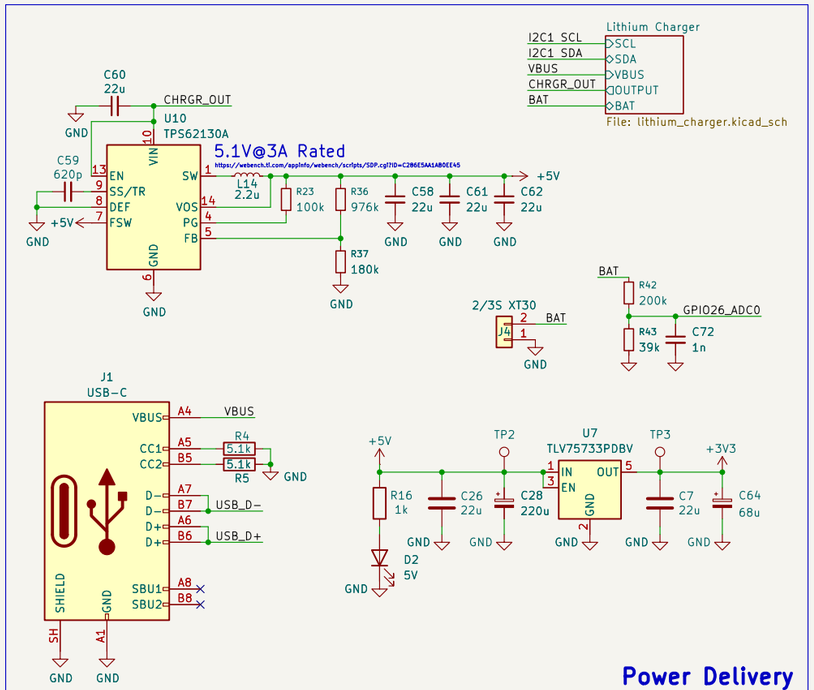
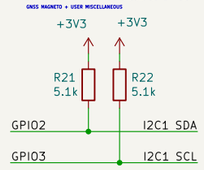
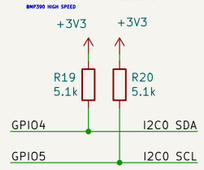
Power Delivery



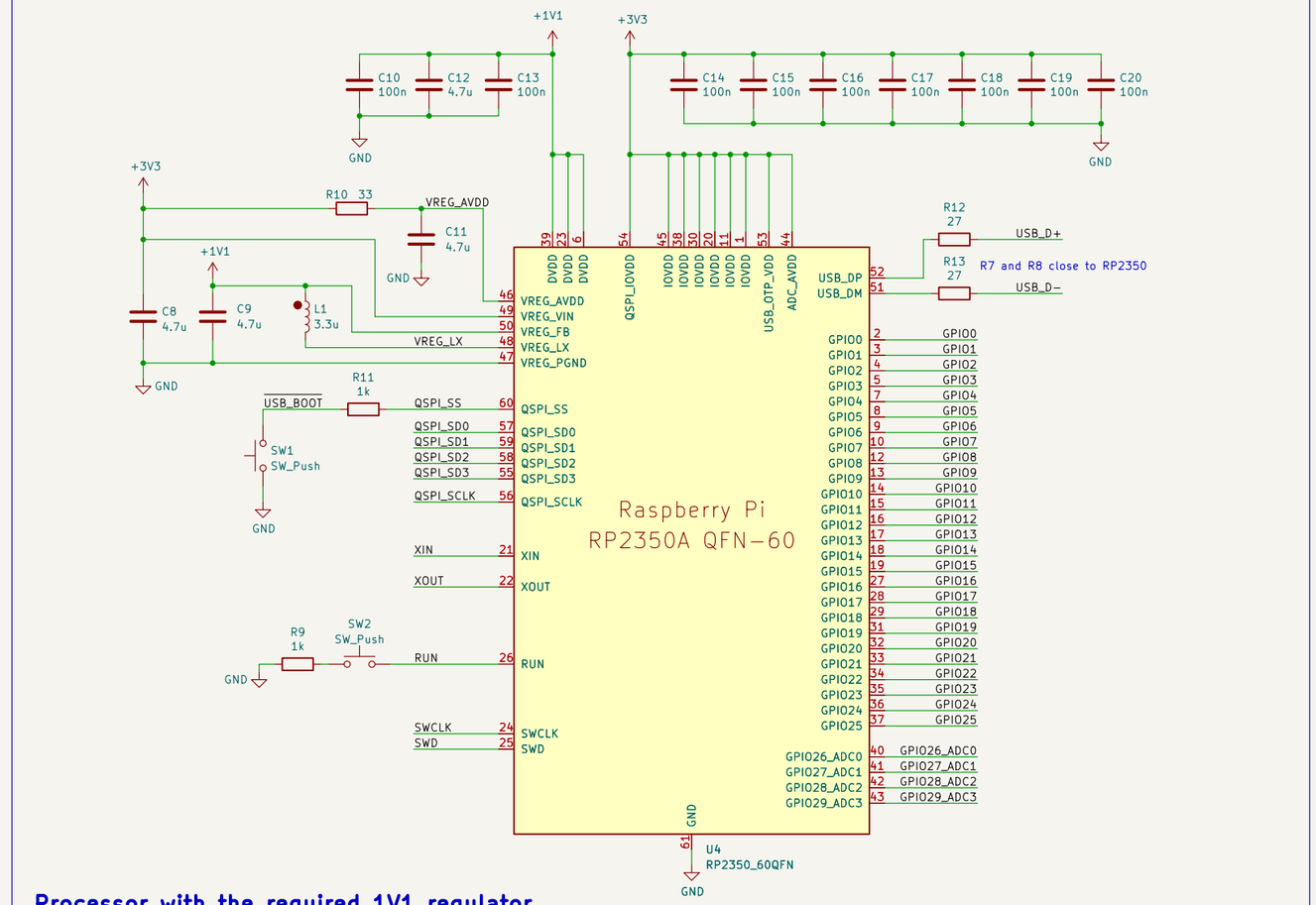
I2C1 PULLUPS



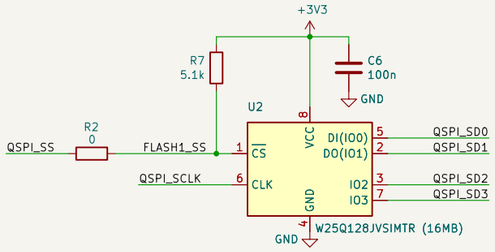
I2C0 PULLUPS



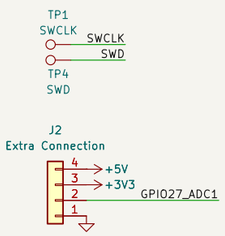
Processor with the required 1V1 regulator



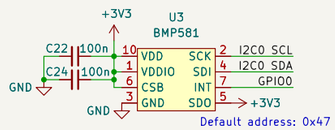
Firmware and Flight Data Flash



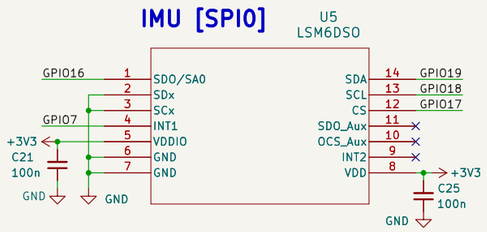
Misc connectors



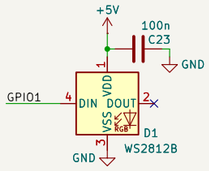
BMP581 [I2C0]



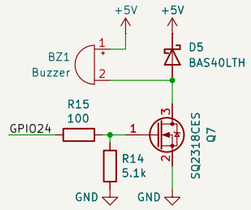
IMU [SPI0]



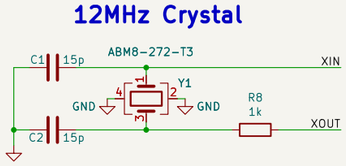
WS2812B



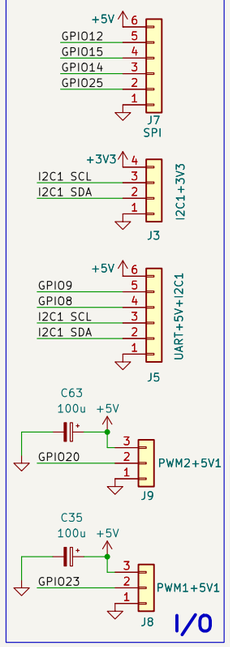
BUZZER



12MHz Crystal



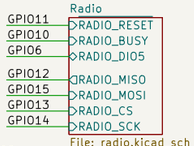
I/O



change notes:

- c32, c31 changed from 100n to 1n (held charge too long)
- Removed Flight Data Flash in favor of single, 128Mbit (16MB) one
- Removed 2.4GHz RF chain
- Replaced BMP390 with BMP581
- Routed GPIO to IMU interrupt output
- Redid power section from battery to account for a 2/3S charger
- All ADCs now use 200k/39k voltage divider

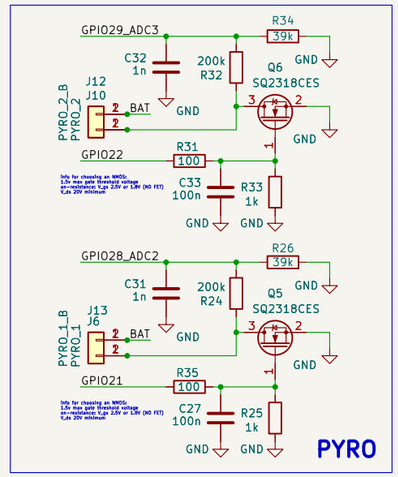
Radio



Mounting



PYRO



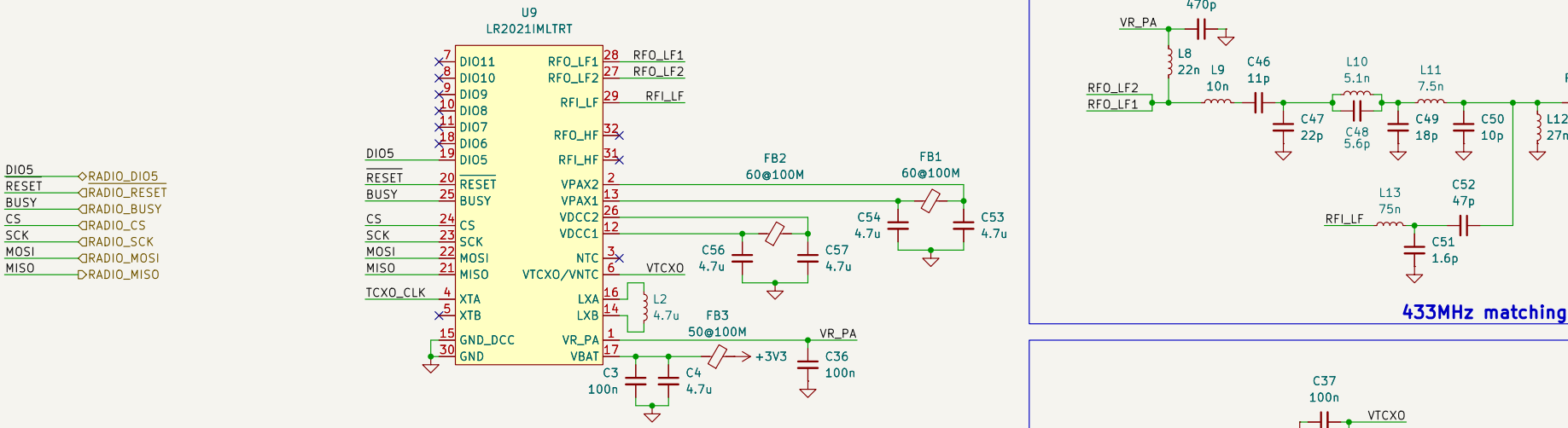
GPIO	FUNCTION
0	BMP interrupt
1	WS2812B SIGNAL
2	I2C1 SDA
3	I2C1 SCL
4	I2C0 SDA
5	I2C0 SCL
6	RADIO DIO5
7	IMU interrupt
8	GNSS UART1 TX
9	GNSS UART1 RX
10	RADIO BUSY
11	RADIO RESET
12	RADIO SPI1 RX [MISO]
13	RADIO SPI1 CSn [CS]
14	RADIO SPI1 SCK [SCL]
15	RADIO SPI1 TX [MOSI]
16	IMU SPI0 RX [MISO]
17	IMU SPI0 CSn [CS]
18	IMU SPI0 SCK [SCL]
19	IMU SPI0 TX [MOSI]
20	PWM2
21	PYRO_1 TRIGGER
22	PYRO_2 TRIGGER
23	PWM1
24	BUZZER TRIGGER
25	External SPI CS
26	BATTERY VOLTAGE ADC
27	Misc Connector
28	PYRO_1 CONTINUITY ADC
29	PYRO_2 CONTINUITY ADC

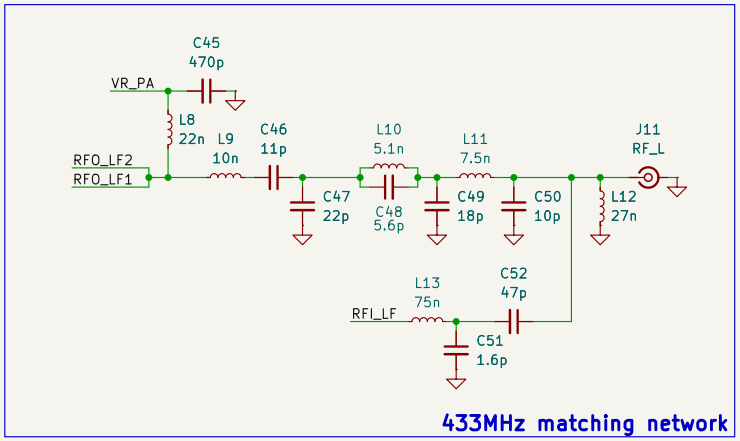
Aria Wiktoria Horak
Silesian Aerospace Technologies

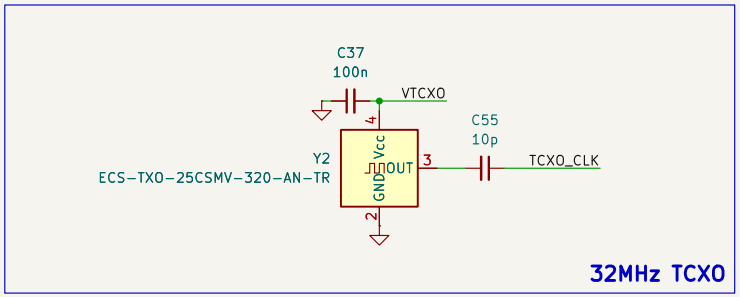
Sheet: /
File: CARP.kicad_sch

Title: CARP-LITE

Size: A3	Date: 2026-09-07	**Rev: V2**
KiCad E.D.A. 10.0.8		Id: 1/3





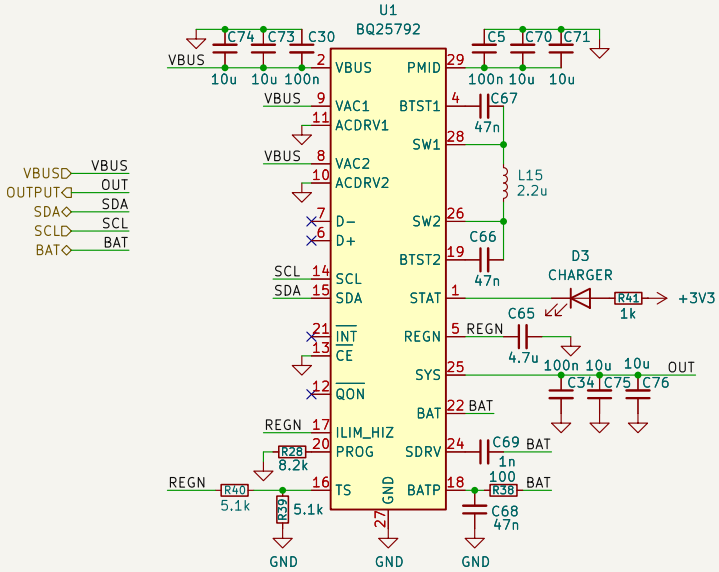


Aria Wiktoria Horak
Silesian Aerospace Technologies

Sheet: /Radio/
File: radio.kicad_sch

Title: CARP-LITE

Size: A4	Date: 2026-09-07	**Rev: V2**
KiCad E.D.A. 10.0.8		Id: 2/3



Aria Wiktoria Horak

Silesian Aerospace Technologies

Sheet: /Lithium Charger/

File: lithium_charger.kicad_sch

Title: CARP−LITE

Size: A4	Date: 2026−09−07	**Rev: V2**
KiCad E.D.A. 10.0.8		Id: 3/3